



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005

ALCOHOL COUNTERMEASURE SYSTEMS CORP.

60 International Blvd.
Toronto, ON Canada M9W 6J2
Charles Esposto Phone: 416 619 3500

CALIBRATION

Valid To: March 31, 2019

Certificate Number: 4169.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Chemical

Parameter/Equipment	Range	CMC ^{2, 4} (±)	Comments
Ethanol Concentration – Measure	(0.1 to 2.4) mg/liter	0.01 mg	NIST SRMs
Ethanol Concentration – Measure (SAF'IR)	(0.1 to 2.0) mg/liter	1.6 %	NIST SRMs

II. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC ^{2, 3} (±)	Comments
DC Voltage – Measure	(0 to 100) mV 100 mV to 1 V (1 to 10) V (10 to 100) V	0.0050 % + 3.5 µV 0.0035 % + 5 µV 0.0040 % + 70 µV 0.0045 % + 0.6 mV	GDM-8261 DMM
DC Current – Measure	100 mA to 1 A (1 to 10) A (10 to 50) A	0.10 % + 0.1 mA 0.15 % + 0.8 mA 0.23 % + 0.21 mA	GDM-8261 DMM, standard resistor

Parameter/Equipment	Range	CMC ^{2, 3} (±)	Comments
DC Current – Generate	(0 to 24) mA	0.01 % + 0.002 mA	Power supply

Parameter/Range	Frequency	CMC ^{2, 3} (±)	Comments
AC Voltage – Measure			GDM-8261 DMM
(1 to 5) mV	10 Hz to 20 kHz	0.06 % + 0.14 mV	
(5 to 100) mV	10 Hz to 20 kHz	0.06 % + 40 µV	

III. Fluid Quantities

Parameter/Equipment	Range	CMC ^{2, 4} (±)	Comments
Flow	(0.5 to 50) SLPM	0.15 %	Flow standard

IV. Mechanical

Parameter/Equipment	Range	CMC ^{2, 4} (±)	Comments
Pressure – Magnehelic Gages	(0 to 10) inH ₂ O	0.5 inH ₂ O	Manometer

V. Thermodynamics

Parameter/Equipment	Range	CMC ² (±)	Comments
Temperature – Generate	(-60 to 100) °C (101 to 150) °C (151 to 325) °C	1.0 °C 0.16 °C 0.60 °C	Temperature oven

Parameter/Equipment	Range	CMC ² (±)	Comments
Temperature – Measure	(0 to 100) °C (-200 to 300) °C (300 to 500) °C	0.01 °C 0.03 °C 0.09 °C	PRT and indicator

VI. Time & Frequency

Parameter/Equipment	Range	CMC ² (±)	Comments
Stopwatches and Timers	0.1 s to 100 hours	0.82 s	NIST course time

¹ This laboratory offers commercial calibration service.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ The stated measured values are determined using the indicated instrument (see Comments). This capability is suitable for the calibration of the devices intended to measure or generate the measured value in the ranges indicated. CMC's are expressed as either a specific value that covers the full range or as a percent or fraction of the reading plus a fixed floor specification.

⁴ In the statement of CMC, percentages represent the percent of a reading unless otherwise noted.



Accredited Laboratory

A2LA has accredited

ALCOHOL COUNTERMEASURE SYSTEMS CORP.

Toronto, Ontario, CANADA

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the R205 – *Specific Requirements: Calibration Laboratory Accreditation Program* requirements. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).



Presented this 22nd day of December 2016.

A handwritten signature in black ink, appearing to read 'L. Sen', written over a horizontal line.

President and CEO
For the Accreditation Council
Certificate Number 4169.01
Valid to March 31, 2019

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.